Lab Task 3

Vehicle Rental Management System

First, Create an Abstract Vehicle Class

First, we create an abstract class called **Vehicle**.

It contains a protected String vehicleId to store the unique ID of each vehicle.

We also create a constructor to initialize the vehicle ID.

Then Create an Abstract Method

We create the method calculateRentalCost(int days).

This method is abstract, so it does not have a body in the Vehicle class. Each subclass provides its own implementation to calculate the rental cost.

We also create printRentalDetails() to display the vehicle ID, rental duration, and total rental cost.

Then Create the Car Class

We create a **Car** class that extends **Vehicle**.

extends means Inheritance.

The **Car** class has an additional property, dailyRate, which stores the rental cost per day.

We override **calculateRentalCost()** to calculate the total cost by multiplying the daily rate by the number of days. If the rental duration is zero or negative, the method returns 0.

Then Create the Bike Class

We create a **Bike** class that also extends **Vehicle**.

It has a dailyRate property and overrides the **calculateRentalCost()** method.

If the rental duration is more than five days, the customer receives a **10% discount**. Invalid rental durations return 0.

Then Create the LuxuryCar Class

We create **LuxuryCar,** which **extends** Car.

It inherits the dailyRate property from Car and adds a new property called insuranceFee.

Its overridden calculateRentalCost() method calculates the daily rental cost and adds the insurance fee. If the rental duration is zero or negative, it returns 0.

Finally, Create the Main Class

In the main() method, we create a **Vehicle[]** array containing different vehicle objects: Car, Bike, and LuxuryCar.

This demonstrates **Runtime Polymorphism**, because the same array can store different types of vehicle objects.

We use a loop to calculate and print each vehicle's rental details. We also include a rental duration of zero to demonstrate input validation.